Literature Review

2.1 Collaborative awareness and teamwork

One of the key things we explored in our research was how students stay aware of what their teammates are doing during group work. Collaboration is not just about being able to work on the same document. It is also about knowing what other people are contributing, understanding their progress and making sure everyone feels included.

Our research draws on Schnaubert's work, which highlights the importance of being aware of teammates' knowledge, activities and progress. This relates to the problems we identified in university group assessments, where students can become unsure about who is doing what, repeat work that someone else has already completed, or feel that the workload is not being shared fairly.

This helped us recognise that simply giving students another shared digital workspace would not necessarily solve these problems. We wanted to explore how the design itself could make collaboration more visible and encourage students to stay involved in the group's work.

For Zook, this influenced our idea of giving each participant their own identity within a shared workspace. Rather than everyone interacting with one shared device through a single set of controls, we wanted to explore how individuals could have their own representation while still working together on the same surface.

2.2 Technology in educational environments

We also considered how technology is used in educational settings and whether it always supports the way students naturally work together. Our project research refers to Crompton's work on the use of technology in education. This encouraged us to think about how technology could be better integrated into classroom activities, rather than simply adding another digital tool for students to use.

In group assessments, students often move between face-to-face discussion, physical brainstorming and online platforms. Each approach has its benefits, but moving between them can create extra work. For example, ideas created on butcher paper might need to be photographed or recreated digitally, while decisions made during a classroom discussion may need to be transferred into a separate document or group chat.

This led us to focus on the connection between physical and digital collaboration. Instead of replacing face-to-face interaction with individual screens, we wanted to explore how technology could support students while they continued to gather around a shared workspace.

For Zook, this meant exploring a digital tabletop where students could work together in person while retaining their work in a digital format. The aim was to make the transition between classroom collaboration and ongoing digital work more connected.

2.3 Playfulness and Tangible Interaction

Another area we explored was the use of physical objects and play in collaborative experiences. We were interested in whether giving students a physical object to represent them could make the experience more engaging and encourage interaction between group members.

Jacobs and Statler (2006) discuss the role of serious play in developing scenarios and exploring ideas. This was relevant to our thinking because we wanted to explore an approach to collaboration that felt more interactive than using a conventional digital interface. Our research notes also identify Fletcher and Thornton (2025) as a source for understanding different forms of play in workplace settings.

These ideas helped inform our decision to experiment with physical figurines. We considered whether the figurines could represent individual students, connect them to their digital identity and introduce a more playful element into group work.

However, we did not want to assume that making an activity playful would automatically improve teamwork. The figurines needed to be useful within the actual process of collaboration, rather than becoming an extra object that students had to think about or manage.

This became particularly important during our workshops. Although participants showed interest in the figurines and interacted with them, having to hold them while contributing was not always convenient. This helped us recognise that the way a tangible object is incorporated into an activity matters just as much as the object itself.

As a result, we began exploring how the figurines could remain part of the shared digital workspace, rather than requiring participants to constantly hold them. We also considered how interactions between figurines could encourage people to engage with one another, rather than only interacting with their own object.

2.4 Shared surfaces and co-located collaboration

We were also interested in how a shared digital surface could support people working together in the same physical space. Our initial research was influenced by Mark Weiser's concept of badges, pads and boards, as well as our exploration of multi-touch tables.

This helped us think about the relationship between individual identity and shared activity. Students need ways to contribute as individuals, but they also need a common space where they can see, discuss and build on each other's ideas.

Existing digital collaboration tools can support shared work, but they are often accessed through individual devices. Physical tools such as butcher paper and whiteboards provide a shared focal point, although the work can be difficult to continue editing after the session ends.

We wanted to explore whether a digital tabletop could bring these benefits together. With Zook, the shared surface provides a common space for group work, while the figurines represent individual participants within that environment.

Our goal was not to make every student work separately, but to support individual contributions within a shared and social experience. This became a central idea behind the Zook concept.

Existing Solutions

We examined existing physical and digital collaboration approaches to understand what they support and where opportunities for improvement remain.

3.1 Physical collaboration tools

Butcher paper and whiteboards
Physical collaboration tools allow participants to gather around a common surface, discuss ideas and contribute to the same visible artefact. They can support spontaneous discussion without requiring students to navigate a digital interface.

However, the work is tied to the physical surface. Once a session finishes, groups may need to photograph, recreate or manually transfer their work to a digital platform. A photograph preserves a record of the work but does not preserve the original artefact as an editable digital workspace.

What this means for Zook: retain the shared, visible nature of physical collaboration while exploring how work can be captured and revisited digitally.

3.2 Digital collaboration platforms

The team considered platforms including Google Docs, Miro, Padlet, Canva, Echopoll and Ripple.

These tools provide different ways to share information, create content, collect responses and collaborate digitally. Shared documents and online workspaces can make information easier to access beyond the classroom.

However, digital collaboration can also be distributed across multiple applications. Students may need to exchange links, invite team members, manage separate documents and communicate through additional messaging platforms. When students are physically together, individual devices can also divide attention between screen-based activity and group discussion.

These limitations do not mean existing platforms are ineffective. Rather, they highlight an opportunity to improve how physical and digital collaboration connect.

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